

Super-Resolution Reconstruction of Ionospheric TEC Through the Spatiotemporal Complementarity of GNSS and NISAR

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1. Background and Objectives

1.1 Background

Total electron content (TEC) is the number of electrons in a column of unit cross-sectional area along a signal path. It is a key measure of ionospheric structure. Variations in ionospheric electron density refract, delay, and disperse radio waves, affecting satellite navigation, communication, and radar. Accurate TEC maps with high spatial and temporal resolution are useful for ionospheric research, radio-propagation correction, GNSS positioning, spacecraft operations, and monitoring space-weather and earthquake-related disturbances.

Global Navigation Satellite Systems (GNSS), including GPS, BeiDou, GLONASS, and Galileo, provide continuous measurements. The ionospheric delay of a GNSS signal is proportional to the electron content along its path and inversely proportional to the square of its frequency. Dual-frequency observations can therefore be used to estimate slant TEC; a thin-shell ionosphere model and a mapping function convert this to vertical TEC. Ground stations are spatially sparse, however, so GNSS-derived TEC fields have limited spatial resolution even though their temporal coverage is nearly continuous.

NISAR (NASA-ISRO Synthetic Aperture Radar) carries L- and S-band radars in an approximately 747 km Sun-synchronous orbit with a nominal 12-day revisit. Its interferometric observations offer much finer spatial sampling. Frequency-dependent ionospheric phase can be separated from other interferometric terms and used to estimate an ionospheric phase field related to TEC. The revisit interval limits temporal coverage at a given location. GNSS and NISAR thus offer complementary temporal and spatial information.

1.2 Objective

This study tests whether a deep-learning model can infer a high-resolution NISAR ionospheric phase map from a processed GNSS TEC-derived phase field. A lightweight residual U-Net takes the GNSS field as input and the corresponding NISAR ionospheric phase as its target. We ask whether the GNSS observations contain enough information to recover fine spatial texture, or only to constrain the broad-scale trend. A negative result is informative because it identifies limits of cross-modal learning under the available sampling conditions.

2. Data Sources and Matching

We obtained 78 NISAR Level-1 raw acquisitions from the Alaska Satellite Facility search portal (<https://search.asf.alaska.edu/>), covering 12 California frames: track 42, frames 69, 70, and 72; track 85, frames 69, 70, 71, and 73; and track 106, frames 17-21. Their acquisition dates span 23 October 2025 to 12 January 2026. These raw data require interferometric processing before they can serve as machine-learning targets.

The corresponding GNSS product came from the Madrigal service (<http://madrigal.iggcas.ac.cn/>): global TEC maps sampled every five minutes on a 1° longitude by 1° latitude grid. We converted TEC differences to ionospheric phase differences using Equation (1), selected the nearest GNSS time for each NISAR acquisition, and cropped the GNSS field to the NISAR footprint.

$$(1) \Delta\phi_{\text{iono}} = -(4\pi K / (c f_0)) \Delta\text{TEC}; K = 40.28 \text{ m}^3/\text{s}^2.$$

Here c is the speed of light and f_0 is the radar carrier frequency. The available processing yielded 28 paired GNSS-NISAR samples for model training and analysis.

3. NISAR Data Processing

The raw NISAR acquisitions were processed into high-resolution ionospheric phase targets. Figure 1 summarizes the workflow.

1. Acquisition	Pair adjacent scenes from the same track and frame; prepare a DEM.
2. Subbands	Split the L-band signal; co-register, form interferograms, remove topography, filter, and unwrap each subband.
3. Separation	Align grids, trim unreliable range edges, and form a split-spectrum ionospheric phase estimate.
4. Filtering	Remove a polynomial trend; iteratively fill gaps and low-pass filter the residual; restore the trend.
5. Output	Return to the original radar grid and apply the observation-validity mask.

Figure 1. NISAR processing workflow, redrawn in English from the original report.

3.1 Interferometric pairs

A single complex SAR acquisition contains phase associated with sensor-to-ground range, but it does not directly isolate ionospheric change. We paired temporally adjacent acquisitions from the same track and frame. The later image was treated as the reference image and the earlier image as the secondary image. Their phase difference formed an interferogram. A digital elevation model (DEM) extending beyond the common footprint was prepared so that topographic phase could be removed during interferometric processing.

3.2 Two-subband interferometric phase

The ionosphere is dispersive: lower-frequency radar signals experience a larger phase effect. Topography, orbital geometry, and ground displacement follow a different frequency dependence. We split the wide L-band signal into lower- and higher-frequency subbands and processed each as an interferometric observation. The workflow included conversion of the raw data, precise co-registration, removal of topographic phase, interferogram formation and filtering, and phase unwrapping. This produced two unwrapped phase maps for the same interferometric pair, with similar nondispersive content but different ionospheric contributions.

3.3 Split-spectrum ionospheric phase estimation

The two subband phases were combined using frequency-dependent weights to separate the dispersive ionospheric term. Before combining them, we resampled the higher-frequency result onto the lower-frequency grid so corresponding pixels represented the same locations. Nearest-neighbor resampling avoided introducing mixed phase values through smoothing. We also trimmed the near- and far-range edges, where phase quality was less reliable. The resulting frequency-weighted difference gave an initial ionospheric phase estimate, but it still contained noise, missing pixels, and large-scale artifacts.

3.4 Filtering and reconstruction of the phase field

The desired ionospheric component was treated as a spatially smooth, long-wavelength field. To set a physically meaningful low-pass cutoff, radar-sample coordinates were converted to ground distances in meters. Missing data made direct frequency-domain filtering unsuitable: zero filling would create sharp boundaries and spectral leakage.

We therefore used iterative gap filling and filtering. First, a low-order polynomial surface was fitted and subtracted to remove the broad trend. The residual was low-pass filtered repeatedly; after each iteration, observed values at valid pixels replaced the filtered values, while the filter filled only missing regions. The fitted trend was then added back. This approach produced a continuous field without unnecessarily altering reliable observations.

Finally, the range edges removed earlier were restored as missing values, and the estimate was resampled to the original high-resolution grid. The validity mask of the higher-frequency unwrapped phase was reapplied. No values were extrapolated into unsupported regions. The result was a filtered NISAR ionospheric phase field on the radar grid. Figure 2 compares this target with the spatially much coarser GNSS-derived field.

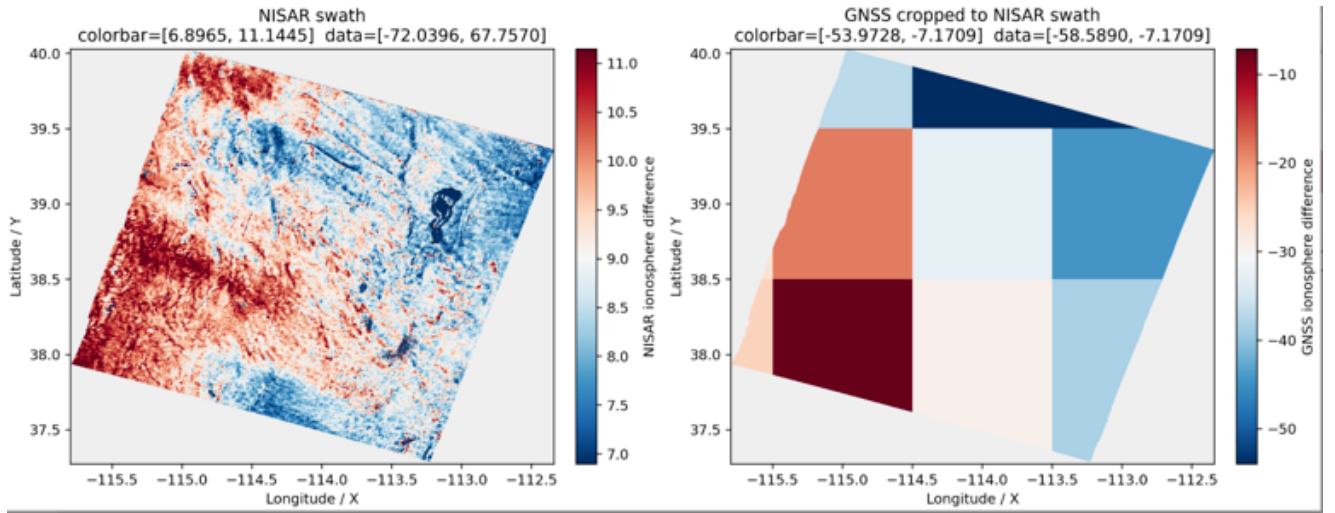


Figure 2. Comparison of a NISAR ionospheric phase estimate (left) and a cropped GNSS-derived field (right).

4. Model Training

4.1 GNSS interpolation

A residual U-Net requires input and target arrays on the same grid. We used kriging to interpolate the GNSS phase field. Kriging determines weights from spatial covariance, commonly represented by a semivariogram, and gives a linear unbiased estimate with minimum estimation variance under its assumptions.

Some NISAR footprints contained no valid GNSS grid points, making direct local interpolation meaningless (Figure 3). We therefore interpolated a larger region, 30°-40° N and 120°-110° W, and cropped the required footprint afterward. Figure 4 shows the large-area interpolation, and Figure 5 compares a cropped interpolation with a NISAR target.

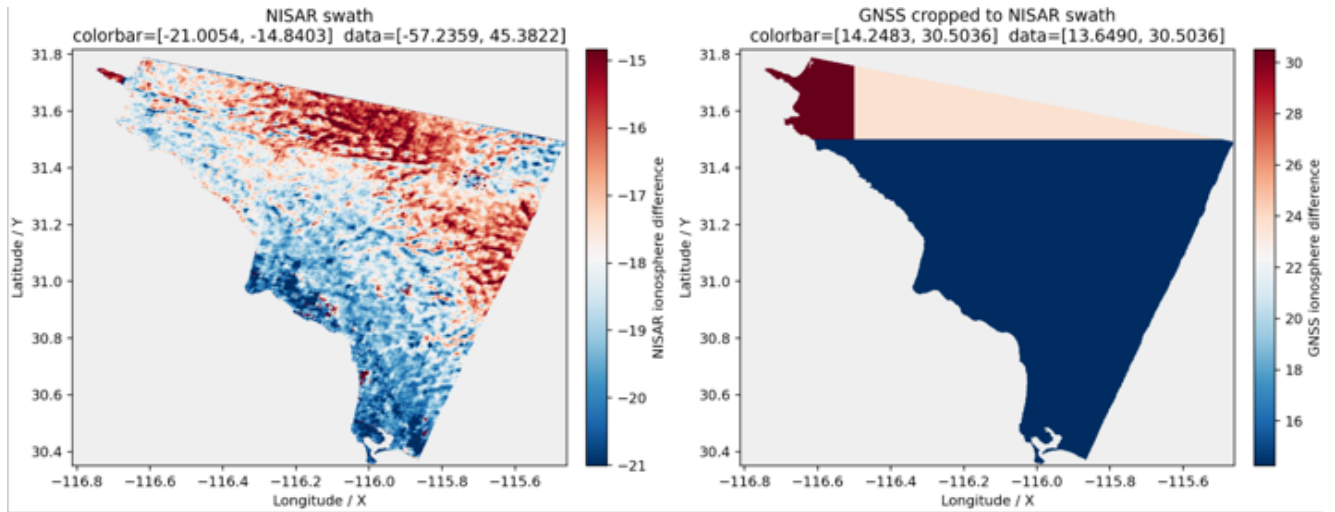


Figure 3. Example of a NISAR footprint with too few valid GNSS points for local interpolation.

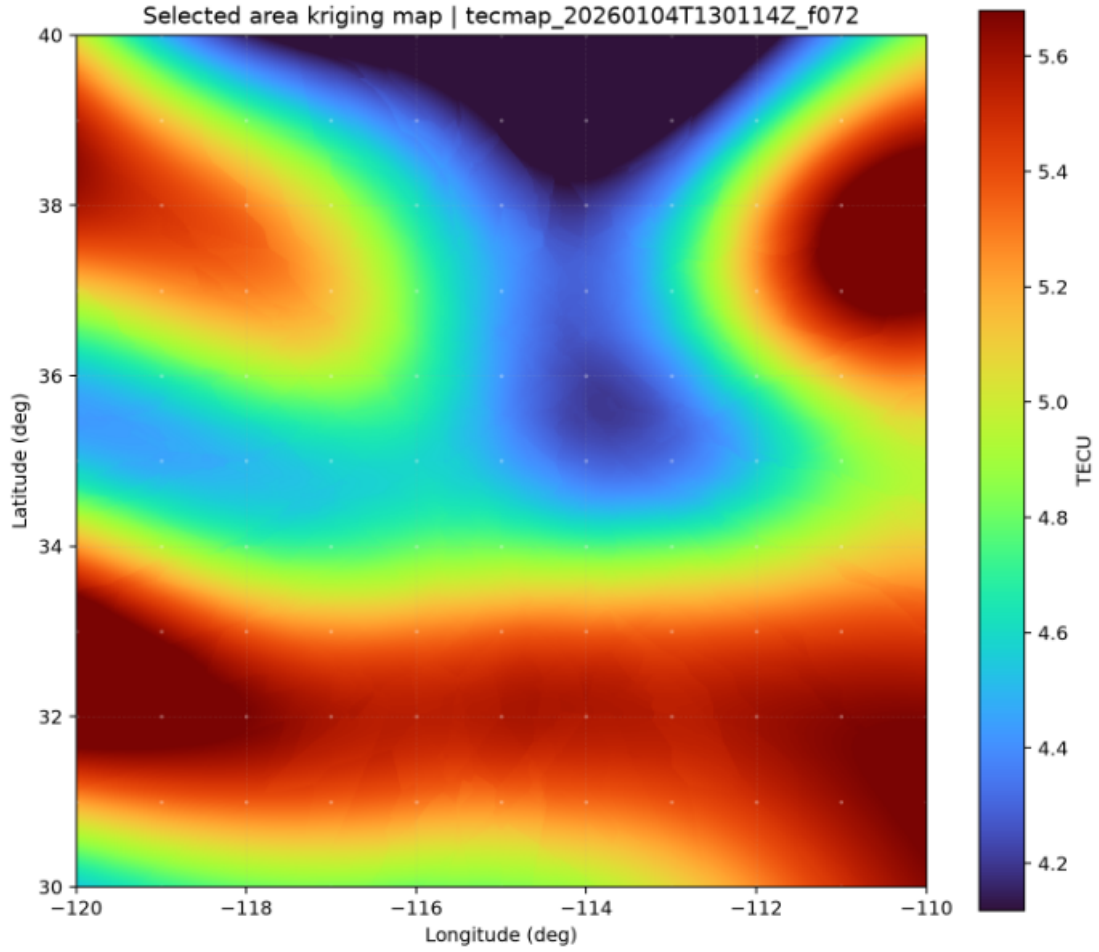


Figure 4. Kriging over the wider California study region before cropping to a NISAR footprint.

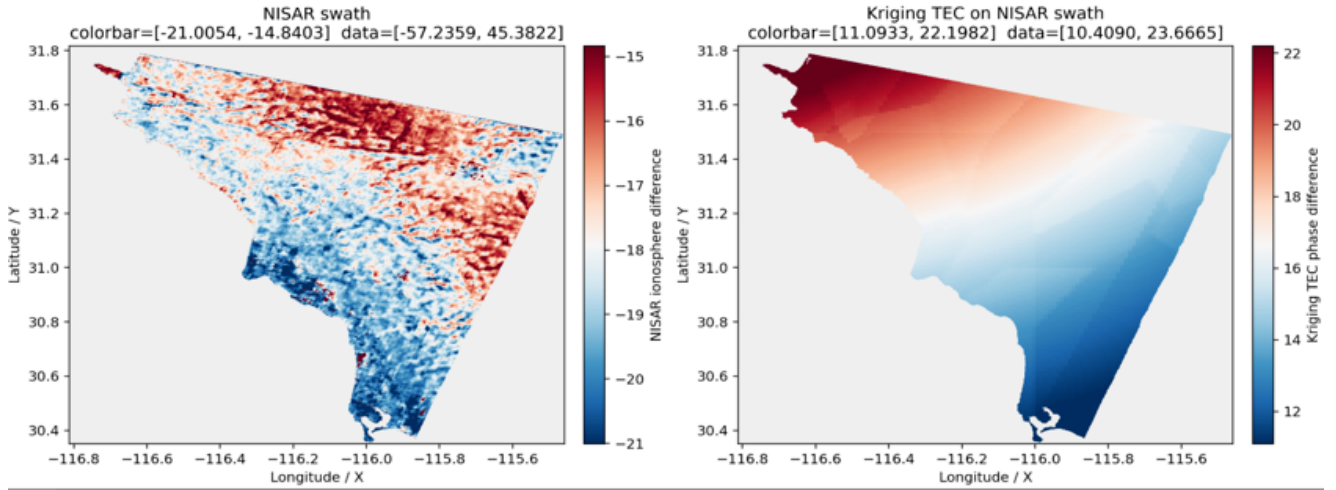


Figure 5. Example of the kriging result within a NISAR swath (right) and the corresponding NISAR phase (left).

4.2 Preprocessing and training strategy

Each training pair contained a GNSS input image and a NISAR target resampled to a common grid. A validity mask limited the loss and evaluation to pixels inside the NISAR swath.

The two phase products have different reference levels: an interferometric NISAR phase has an arbitrary additive constant, whereas the GNSS-derived phase has an absolute reference. We subtracted the mean over valid pixels from each image so that training emphasized spatial variation rather than a sample-specific offset. At inference, the GNSS reference was used to restore an absolute level.

NISAR images were approximately 4000×4000 pixels, but only 28 paired samples were available. Full-image training would have required substantial GPU memory and would have increased overfitting risk. We instead sampled 200 random 128×128 patches per image. Random rotations and flips augmented the patches, and crop locations were resampled each epoch.

4.3 Architecture, losses, and full-image inference

The residual U-Net predicted a correction to the demeaned GNSS input rather than generating the target field from scratch. With G' denoting the demeaned GNSS field and F_{θ} the learned residual function, the demeaned NISAR prediction was

$$(2) N_{\text{pred}}' = G' + F_{\theta}(G').$$

This residual form allows the output to remain close to the input when the data do not support a reliable high-frequency correction.

The training objective combined three terms. A masked L1 loss measured the absolute phase error only at valid pixels (Equation 3). A gradient loss penalized differences in horizontal and vertical spatial gradients (Equation 4). A scalar offset loss constrained the predicted difference between the GNSS and NISAR means (Equation 5). Here M_i is the binary validity mask, μ_G and μ_N are the respective means, and dmu_{pred} is the predicted offset.

$$(3) L_{1,\text{masked}} = [\sum_i M_i |N_{\text{pred},i}' - N_i'|] / [\sum_i M_i].$$

$$(4) L_{\text{grad}} = \|\nabla_x N_{\text{pred}}' - \nabla_x N'\|_1 + \|\nabla_y N_{\text{pred}}' - \nabla_y N'\|_1.$$

$$(5) L_{\text{offset}} = (\Delta\mu_{\text{pred}} - \Delta\mu)^2; \Delta\mu = \mu_N - \mu_G.$$

The weighted sum of these losses was optimized with Adam, weight decay, adaptive learning-rate adjustment, and early stopping.

At full-image inference, overlapping windows were predicted independently. A two-dimensional Hann window gave higher weight to each patch center and lower weight to its edges. For a pixel covered by several windows, the final value was the weighted average in Equation (6), which reduced seam artifacts.

$$(6) N_{\text{pred}}'(x,y) = [\sum_k w_k(x,y) N_{\text{pred},k}'(x,y)] / [\sum_k w_k(x,y)].$$

Here $w_k(x,y)$ is the Hann weight and $N_{\text{pred},k}'(x,y)$ is the prediction from window k .

5. Results and Interpretation

5.1 Training results

Figures 6 and 7 show representative GNSS inputs, NISAR targets, and model predictions. The model did not recover the fine-scale ionospheric structures visible in the NISAR targets. Its outputs largely followed the broad GNSS trends and contained some stripe-like noise. The intended super-resolution reconstruction was therefore not achieved with this dataset and training setup.

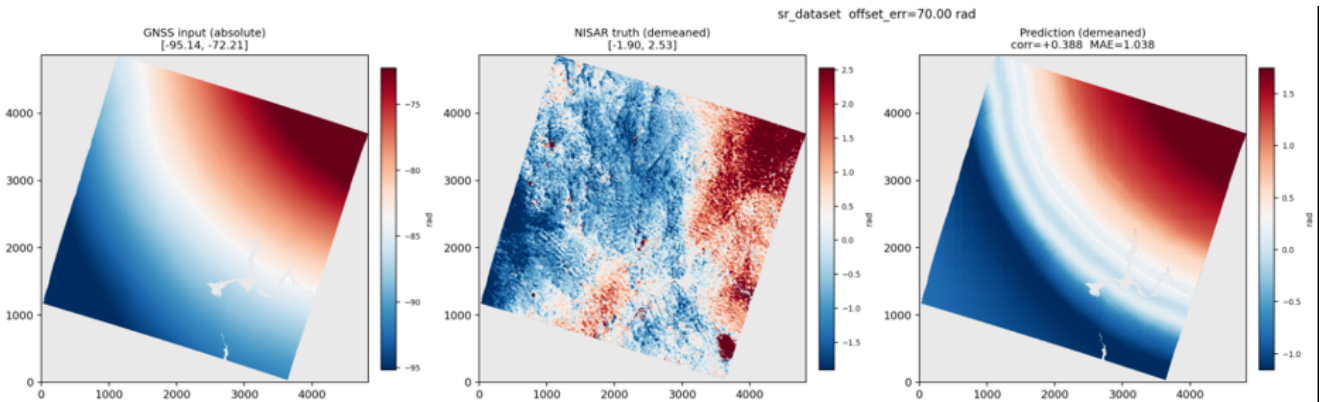


Figure 6. Training example 1: GNSS input, NISAR target, and residual U-Net prediction.

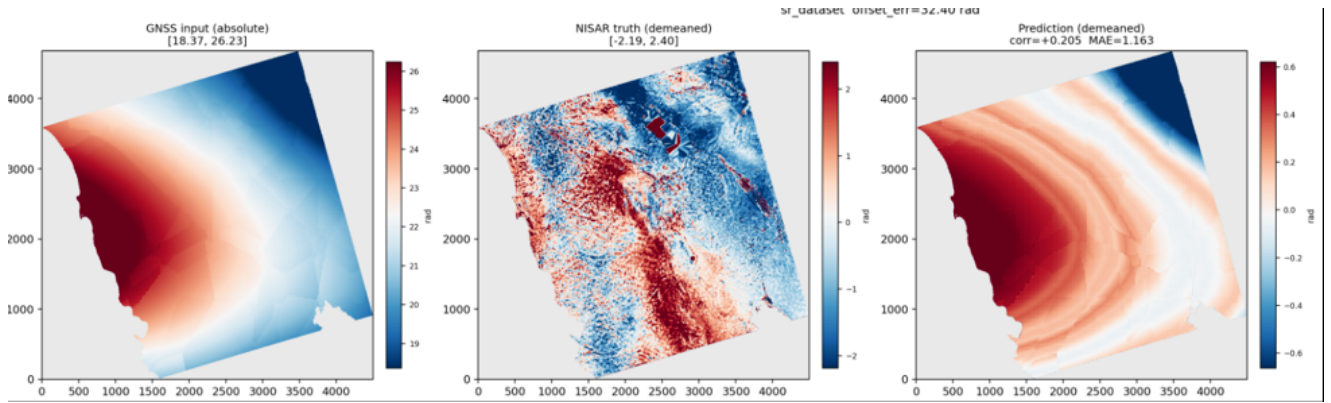


Figure 7. Training example 2: GNSS input, NISAR target, and residual U-Net prediction.

5.2 Why the reconstruction failed

5.2.1 Large resolution and information gap

The main limitation was the scale difference: the GNSS maps resolve structures on roughly 100 km scales, whereas the NISAR phase images can contain features on approximately 100 m scales. Kriging expanded sparse GNSS values into a dense array, but it did not create independent measurements. Each paired sample contained only about eight effective GNSS data points, while its NISAR target contained millions of spatial values. The input thus lacked enough independent information to specify the target's fine texture.

Only 28 paired examples were available to constrain the many possible mappings from sparse, low-dimensional inputs to high-dimensional images. Under these conditions, the residual network tended toward the simpler mapping: preserve the smooth input and avoid unsupported high-frequency predictions. The report therefore attributes the outcome primarily to limited input information and sample count, rather than to an isolated model-architecture or optimization failure.

5.2.2 Mismatched broad-scale trends

Some GNSS and NISAR pairs did not even share the same broad-scale trend (Figure 8), making fine-scale learning less stable. We treated this as a secondary limitation. Restricting training to the eight samples with higher cross-modal correlation still did not yield an effective reconstruction (Figure 9).

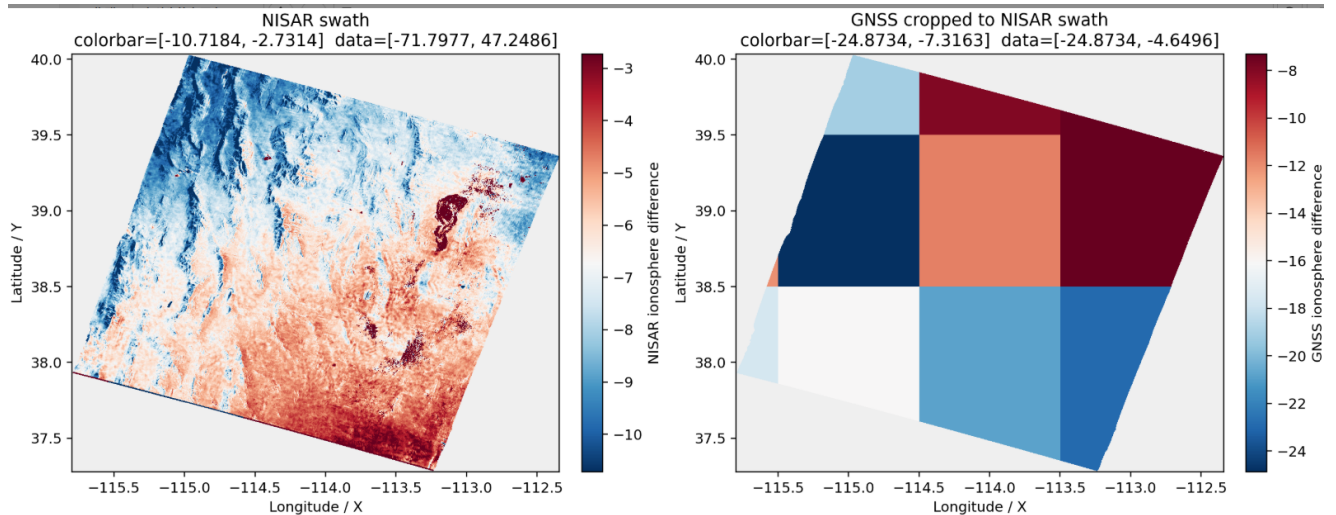


Figure 8. Example of a broad-scale trend mismatch between NISAR and GNSS fields.

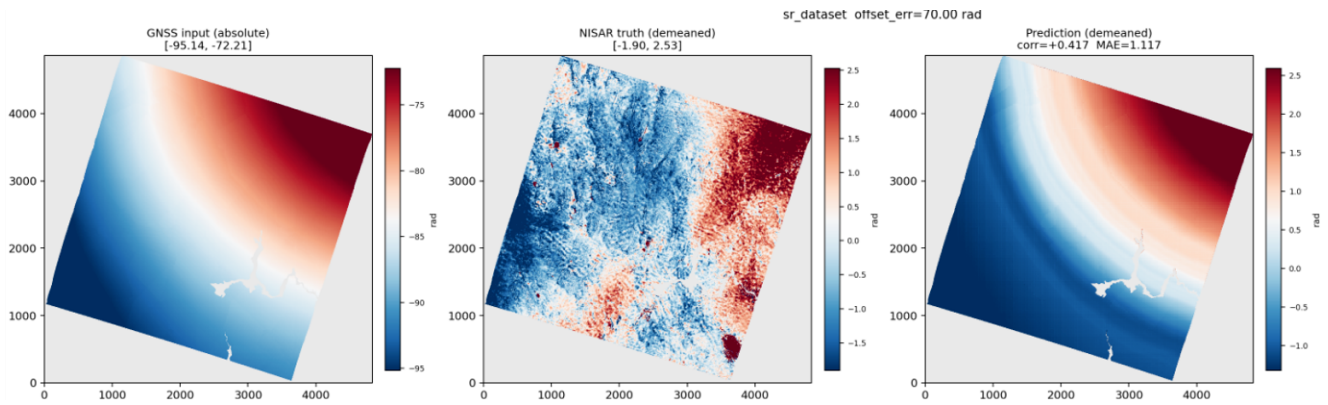


Figure 9. Result after selecting the higher-correlation training samples; fine-scale reconstruction remains unsuccessful.

Two mechanisms may contribute to the trend mismatch. First, the GNSS product is sampled every five minutes, so the nearest match to a NISAR acquisition may differ by up to about 2.5 minutes; ionospheric conditions can change over that interval. Second, the observations have different vertical sensitivity. GNSS signals from satellites at approximately 19,000-24,000 km include plasmaspheric electrons above the ionosphere, whereas a NISAR radar path reaches the spacecraft altitude of about 747 km. Prior work cited in the original report estimated that the plasmasphere can contribute roughly 40%-50% of TEC at night and about 10% by day. The selected acquisitions occurred around 7:00-8:00 p.m. local time, when transport and recombination near dusk may add variability.

5.3 Possible improvements: denser GNSS sampling

A region with many more ground stations could support TEC estimates with more independent spatial information. Processing the raw GNSS observations from such a network, rather than relying only on the coarse gridded product, may narrow the information gap.

5.4 Possible improvements: matching the sampled electron column

The original project discussion proposed correcting the GNSS measurements so that the GNSS and NISAR products represent more comparable electron columns. The IRI-Plas model extends the International Reference Ionosphere from the upper ionosphere to GNSS orbital altitudes. With an electron-density profile informed by GNSS TEC and solar-activity indices such as F10.7 (available from NASA OMNI, <https://omniweb.gsfc.nasa.gov/>), one could estimate the contribution above the NISAR orbit and subtract it from the GNSS column. Equivalently, the profile could be integrated from the ground to approximately 747 km for comparison with the NISAR path. Li et al. (2026) was identified as a methodological reference for TEC calibration. Selecting daytime observations may also reduce the relative plasmaspheric contribution. These were proposed next steps, not validated results of this project.

References

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